

INTERNSHIP REPORT

Dass Scientific Research Labs Pvt. Ltd.

Intern: Porumamilla Akash

Duration: June 2026 – July 2026

Organisation: Dass Scientific Research Labs Pvt. Ltd.

Domain: Software Testing, Quality Assurance (QA), and Application Validation

Mode: Remote

Supervisor: Chanchal Das Sir

* Introduction

I completed a two-month internship at Dass Scientific Research Labs Pvt. Ltd. as a Software Testing Intern. During the internship, I worked on testing and validating DassBridge, DassMath, DassSolitaire, and DassLudo across Android and Windows platforms.

My responsibilities included functional testing, regression testing, integration testing, usability testing, multiplayer validation, defect reporting, and re-testing updated builds. I also tested Android beta releases distributed through the Google Play Store and Windows builds provided by the development team.

* Objectives

- Test the major functionality of Dass applications on Android and Windows.
- Validate complete organiser and player workflows.
- Identify and document reproducible defects with screenshots and clear steps.
- Test multiplayer synchronisation, reconnect handling, scoring, and stability.
- Re-test updated builds and verify whether reported defects were resolved.
- Suggest practical improvements for usability and reliability.

* Applications Tested

DassBridge: Tested tournament creation, registration, seat allocation, player joining, gameplay, claims, scoring, round progression, bot takeover, reconnection, and organiser-player synchronisation.

DassMath: Tested login, profile management, chat, message timestamps, date grouping, file uploads and downloads, course files, update prompts, and chat invitations.

DassSolitaire: Tested single-player and tournament modes, scoring, move count, timer, Undo functionality, card movement, result messages, and round progression.

DassLudo: Tested multiplayer and single-player modes, event joining, dice behaviour, token movement, bot behaviour, completed-token handling, Past Events, and result pages.

* Work Performed

I regularly installed and tested the latest Android and Windows builds. Organiser-created events were joined using multiple devices to verify complete workflows from registration to gameplay completion.

Multiplayer testing was performed with team members to compare organiser and player states across Android and Windows. Bugs were reproduced multiple times and supported with screenshots or videos whenever required.

Each issue was documented with the application name, steps to reproduce, expected behaviour, actual behaviour, and suggestions. Previously reported defects were re-tested after updates to confirm whether they had been resolved.

* Testing Methodology

- Installed the latest Android or Windows application build.
- Created or joined organiser-generated events.
- Executed complete workflows across multiple devices.
- Compared expected and actual application behaviour.
- Reproduced defects multiple times for confirmation.
- Captured screenshots and videos as testing evidence.
- Reported defects to the supervisor and development team.
- Re-tested corrected builds for regression verification.

* Major Issues Identified

- Organiser-created events sometimes failed to connect after successful registration.
- Android applications occasionally remained on a blank, blue, or loading screen after selecting Join Event.
- Tournament gameplay sometimes froze during a player's turn.
- Organiser and player applications displayed inconsistent waiting and connection states.
- Disconnected players were sometimes unable to reclaim their original seat because it appeared occupied.
- Bots occasionally continued playing after the original player had reconnected.
- Bot control was sometimes active without displaying the required Bot indicator.
- Games occasionally displayed "Event unexpectedly ended" during active gameplay.
- Multiplayer applications sometimes crashed or closed during later rounds.
- The Solitaire Undo function reversed card movement but did not correctly restore score and move count.
- Win, Timeout, Out of Moves, and unexpected termination messages were not clearly differentiated.
- The Ludo turn arrow overlapped the dice and reduced visibility.
- Completed or invalid Ludo tokens remained highlighted and selectable.
- Ludo Past Events and Results pages sometimes displayed blank or undefined information.
- DassMath messages from the same day were incorrectly grouped under "Yesterday."
- DassMath course files sometimes downloaded successfully but could not be opened.
- Mobile admin pages were not fully responsive on smaller screens.
- Some Windows installers displayed an "Unknown Publisher" security warning.

* Suggestions Provided

- Improve real-time synchronisation between organiser and player applications.
- Allow successfully reconnected players to reclaim control from temporary bots.
- Display clear Connected, Reconnecting, Disconnected, and Bot Playing states.
- Show notifications when bot takeover or player reconnection occurs.
- Add proper timeout handling instead of indefinite loading screens.
- Improve error messages by displaying the exact cause of failure.
- Correct score and move restoration in Solitaire Undo functionality.
- Display separate result messages for different game-ending conditions.
- Disable selection of invalid and completed Ludo tokens.
- Fix Past Events and Results data retrieval after application reinstallation.
- Correct timezone and date-grouping logic in DassMath chat.
- Digitally sign Windows installers to improve user trust.

* Regression and Beta Testing

Android beta builds were installed and updated directly through the Google Play Store testing programme. Windows builds were also tested for installation, synchronisation, gameplay, and usability.

After each update, previously reported defects were re-tested to determine whether the fixes worked correctly and whether new regressions had been introduced. This process helped assess the stability and release readiness of the applications.

* Documentation and Collaboration

Testing progress was recorded in a daily matrix containing the date, application, issues identified, and suggestions. Screenshots and videos were attached whenever they helped explain loading, synchronisation, UI, gameplay, or result-related defects.

I collaborated with the supervisor, developers, organisers, and other testers. Team testing allowed the same multiplayer event to be examined simultaneously from organiser, Android player, and Windows player perspectives.

* Skills and Learning Outcomes

- Functional, regression, integration, and usability testing
- Android beta and Windows application testing
- Multiplayer and cross-platform validation
- Bug reproduction, documentation, and defect reporting
- Gameplay, scoring, synchronisation, and reconnection testing
- Collection of testing evidence using screenshots and videos
- Communication, collaboration, and release validation

The internship taught me that successful testing requires more than verifying whether a feature works once. A reliable application must behave consistently across devices, users, network conditions, and repeated usage.

I also learned to distinguish intended behaviour from actual defects and to communicate findings clearly enough for developers to reproduce them quickly.

* **Conclusion**

The internship provided practical experience in software testing and quality assurance across four real applications. I tested Android and Windows builds, participated in multiplayer sessions, reported reproducible defects, validated fixes, and suggested improvements to application functionality and user experience.

This experience strengthened my analytical thinking, attention to detail, technical communication, teamwork, and understanding of real-world software release processes.

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